SOC Incident Response Report

Case IR-2026-001 | Microsoft Defender for Endpoint + MySQL Audit | Investigation host: corp-ra01-te123 / corp-ra01-te123da

High Severity — Confirmed Destructive Activity

1. Executive Summary

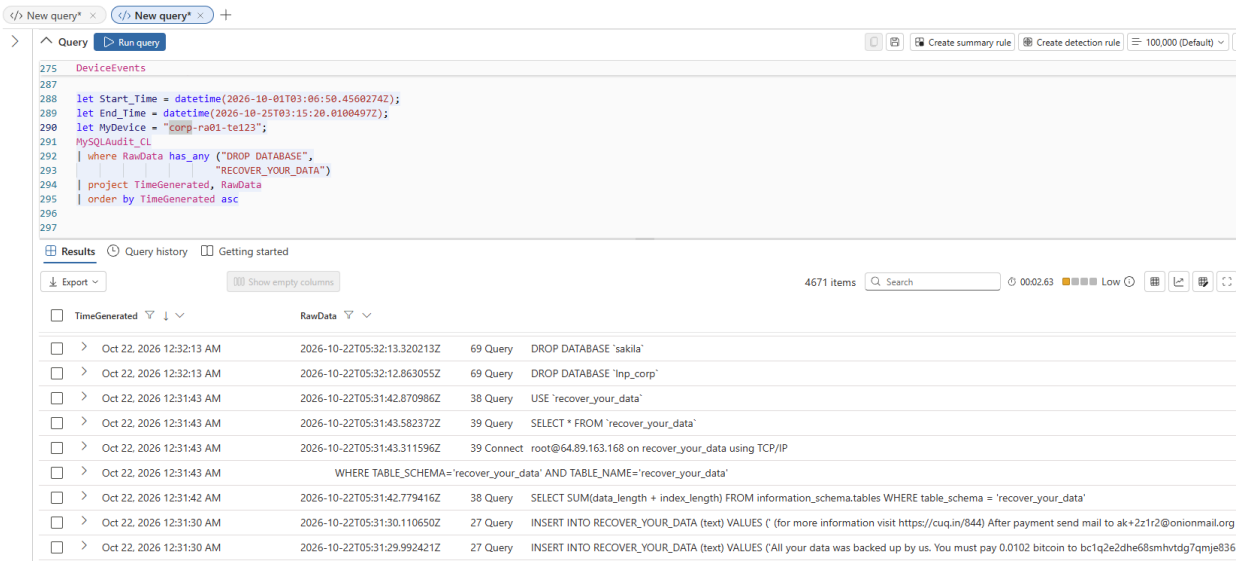
On October 1, 2026, MySQL audit logs recorded suspicious privileged database activity involving ransom-note creation, queries targeting sensitive tables, and commands to delete four databases on `corp-ra01-te123da`. Microsoft Defender telemetry also documented PowerShell execution of `pwncrypt.ps1` and suspicious file operations on `corp-ra01-te123`; however, the relationship between these activities and the database compromise has not been conclusively established. Authentication and network records identify multiple external IP addresses that have communicated with or attempted to access the environment, but the source responsible for the destructive SQL session remains unconfirmed. The incident is assessed as **High severity**, with confirmed destructive SQL activity and extortion indicators; successful data destruction, exfiltration, initial access, and recovery status remain undetermined from available logs.

KQL hunt — identify destructive database activity

```
MySQLAudit_CL
| where RawData contains "DROP DATABASE"
    or RawData contains "RECOVER_YOUR_DATA"
| project TimeGenerated, RawData
| order by TimeGenerated asc
```

Run in Log Analytics only if the MySQL audit data has been ingested into `MySQLAudit_CL`. This is not a native MDE Advanced Hunting table.

Advanced hunting



2. Incident Details

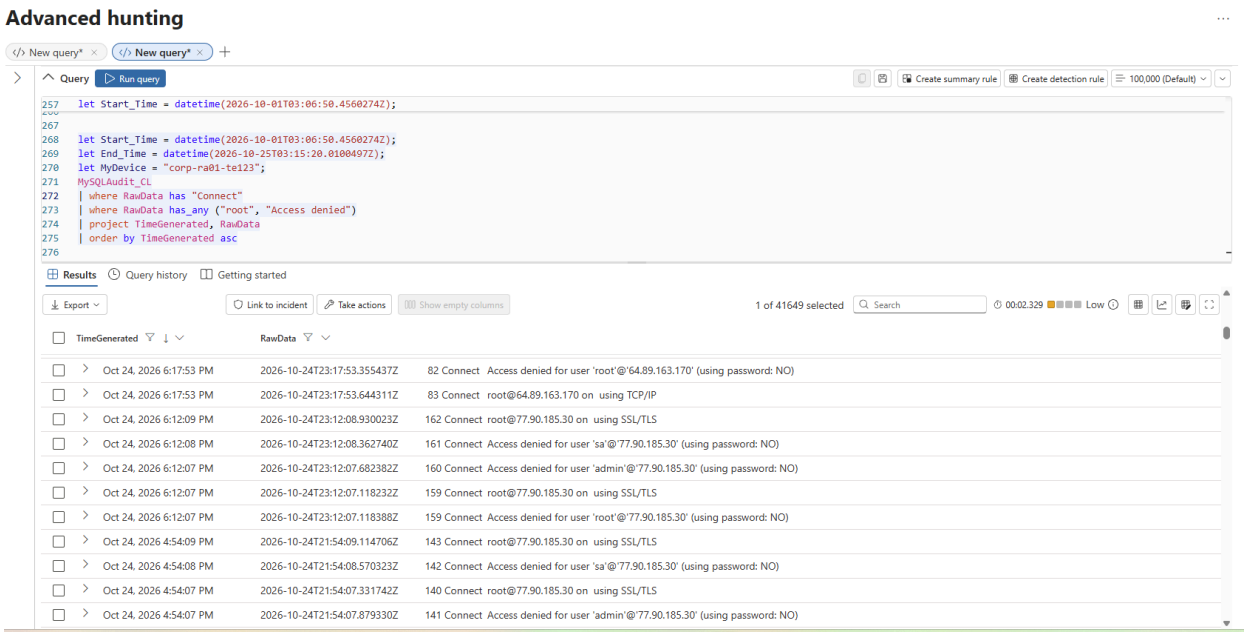
Field	Finding
Classification	Destructive database compromise and extortion; suspicious endpoint file activity
Severity	High
Incident activity	September 30–October 1, 2026, with related activity requiring further correlation
Detection date/time	Not determined from available logs
Reporter	Not determined from available logs

Endpoint	`corp-ra01-te123`
Database host	`corp-ra01-te123da`
MySQL account	`root`
Windows account	`administrator`; other process account contexts require correlation
Affected databases	`cr_corp_01`, `sakila`, `world`, `recover_your_data`

The two hostnames should be correlated in the asset inventory before treating them as a single confirmed asset.

KQL hunt — host logon activity

```
DeviceLogonEvents
| where DeviceName startswith "corp-ra01-te123"
| project Timestamp, AccountName, LogonType,
        ActionType, RemoteIP
| order by Timestamp asc
```



3. Impact Assessment

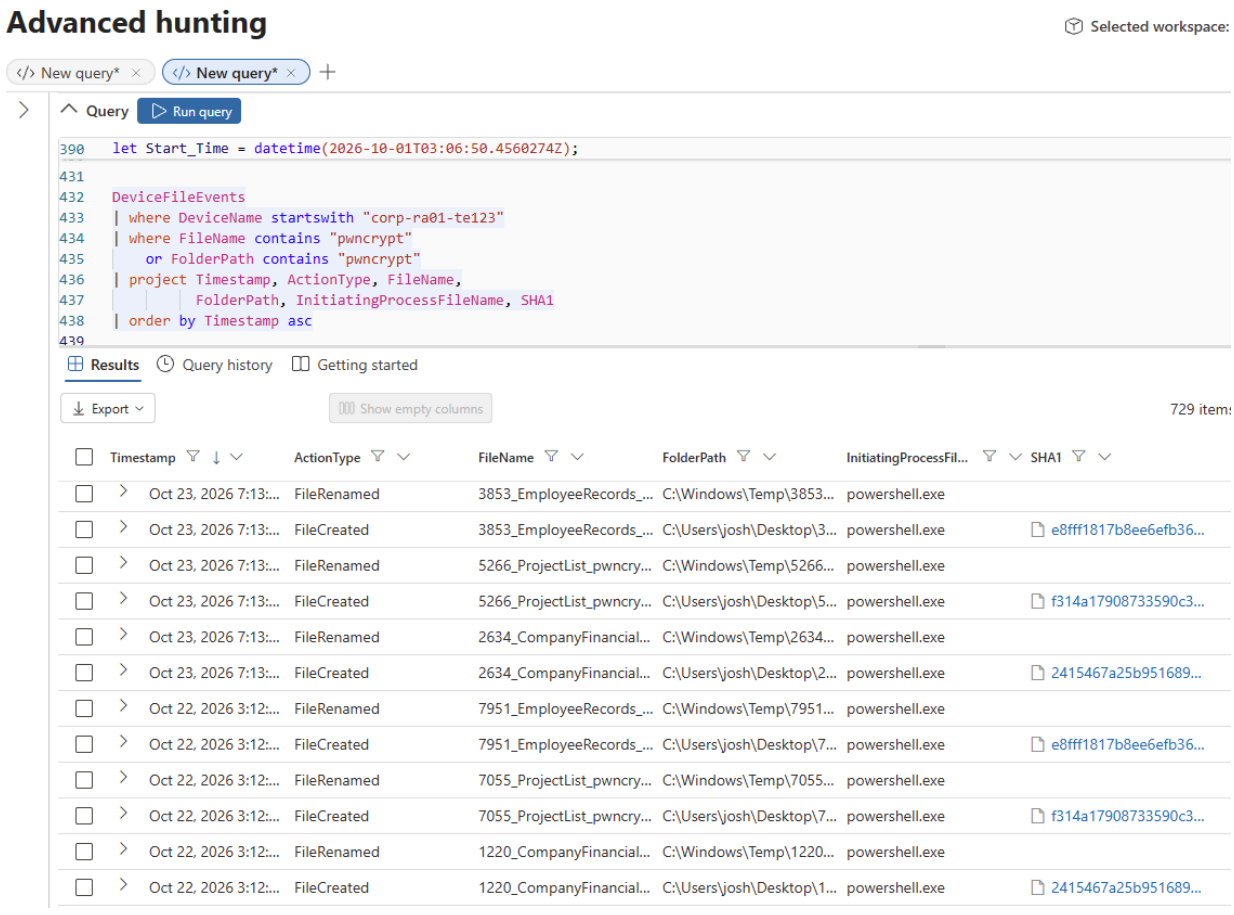
Security objective	Evidence-based assessment
Confidentiality	`SELECT` statements reference `credentials`, `customers`, `orders`, and `payments`. Exfiltration is not determined from available logs.
Integrity	Four `DROP DATABASE` commands were recorded. Endpoint file creation/renaming also warrants investigation.
Availability	Destructive SQL commands and suspicious file operations indicate potential disruption. Actual service outage is not established.
Scope	Four targeted database names; multiple endpoint files with `_pwncrypt.csv` suffixes.

Business impact	Potential database loss and disruption involving financial and employee-related files. Actual loss and recovery costs are unknown.

KQL hunt — suspicious endpoint file changes

```
DeviceFileEvents
| where DeviceName startswith "corp-ra01-te123"
| where FileName contains "pwncrypt"
    or FolderPath contains "pwncrypt"
| project Timestamp, ActionType, FileName,
          FolderPath, InitiatingProcessFileName, SHA1
| order by Timestamp asc
```

Evidence screenshot 1 — Endpoint file modifications



4. Indicators of Compromise

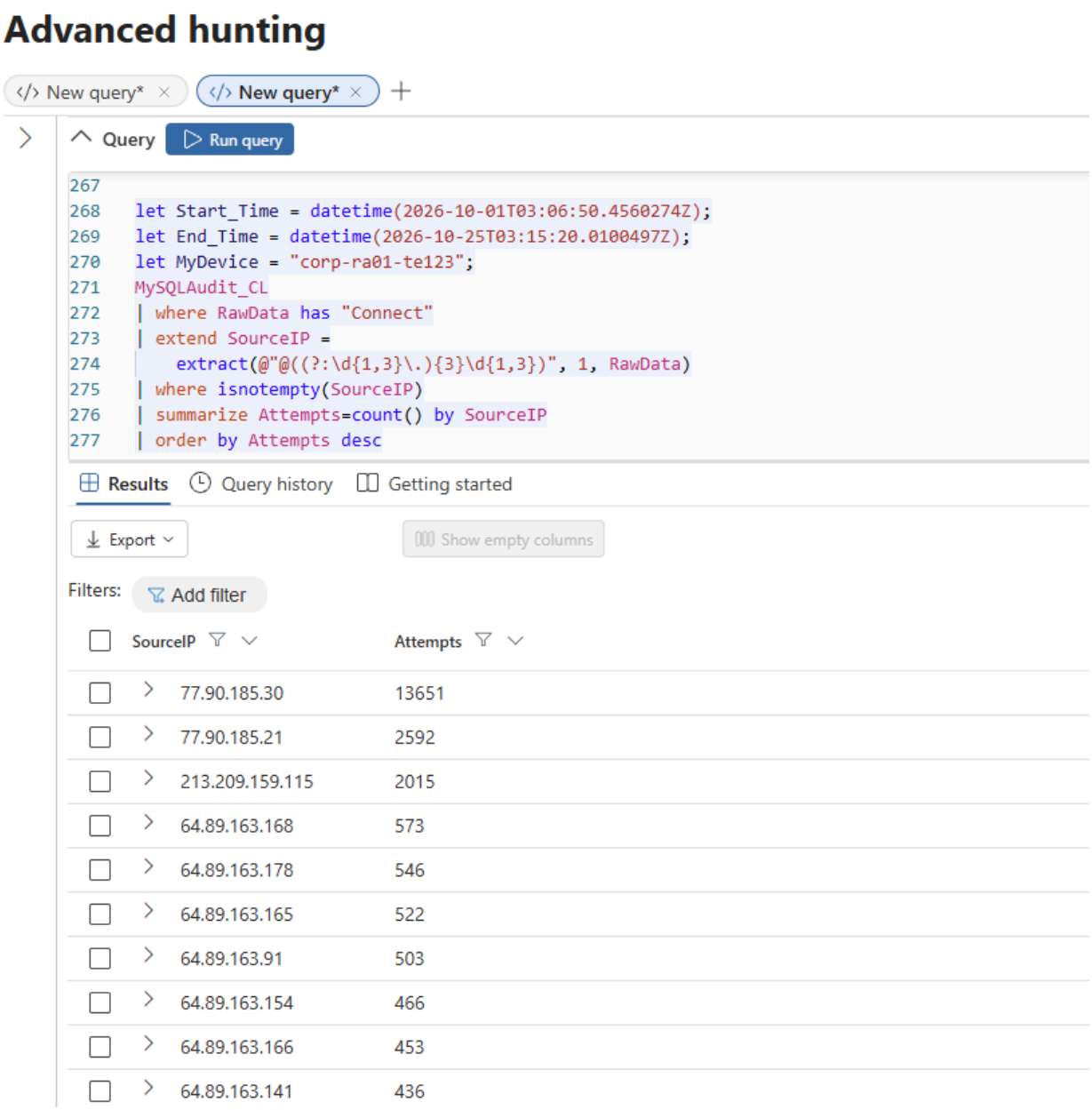
The following indicators are separated into observed evidence and user-supplied indicators that are not found in the SQL exports.

Indicator	Value	Status
Bitcoin address	`bc1qt6kstjjdatqnxd3ksdcfv0jnhdjzndcnkx8zla`	Observed in SQL ransom note
Contact email	`ak+2o2bg@onionmail.org`	Observed in SQL ransom note
Reference URL	`hxxps://cuq[.]in/844`	Observed in SQL ransom note
DATAID	`2O2BG`	Observed
Ransom table	`RECOVER_YOUR_DATA`	Observed
Script	`C:\ProgramData\pwncrypt.ps1`	Observed
File suffix	`_pwncrypt.csv`	Observed
Source IP	`77.90.185.21`	118 authentication records, primarily failures
Source IP	`213.209.159.115`	90 authentication records, primarily failures

Source IP	`77.90.185.30`	63 failures and 4 successes
Windows logon IP	`182.42.225.34`	17 successful logon records
Supplied Bitcoin address	`bc1qk9kvwhzt60u3eqcjllqlj44h0tj7w7n72apz99`	Not found in uploaded SQL exports
Supplied email	`ak+28t2@onionmail.org`	Not found in uploaded SQL exports
Supplied URL	`hxxps://2no[.]co/2mysql`	Not found in uploaded SQL exports
Supplied DATAID	`28T2`	Not found in uploaded SQL exports

KQL hunt — investigate network indicators

```
DeviceNetworkEvents
| where DeviceName startswith "corp-ra01-te123"
| where RemoteIP in (
   "77.90.185.21",
   "77.90.185.30",
   "213.209.159.115",
   "182.42.225.34")
   or LocalPort == 3306
| project Timestamp, ActionType, LocalPort,
        RemoteIP, RemotePort,
        InitiatingProcessFileName
| order by Timestamp asc
```



5. Incident Timeline

MySQL event times below are UTC, taken from the original `RawData` timestamps. Endpoint and network CSVs contain formatted display times without explicit offsets, so their precise ordering against UTC database events requires timezone normalization.

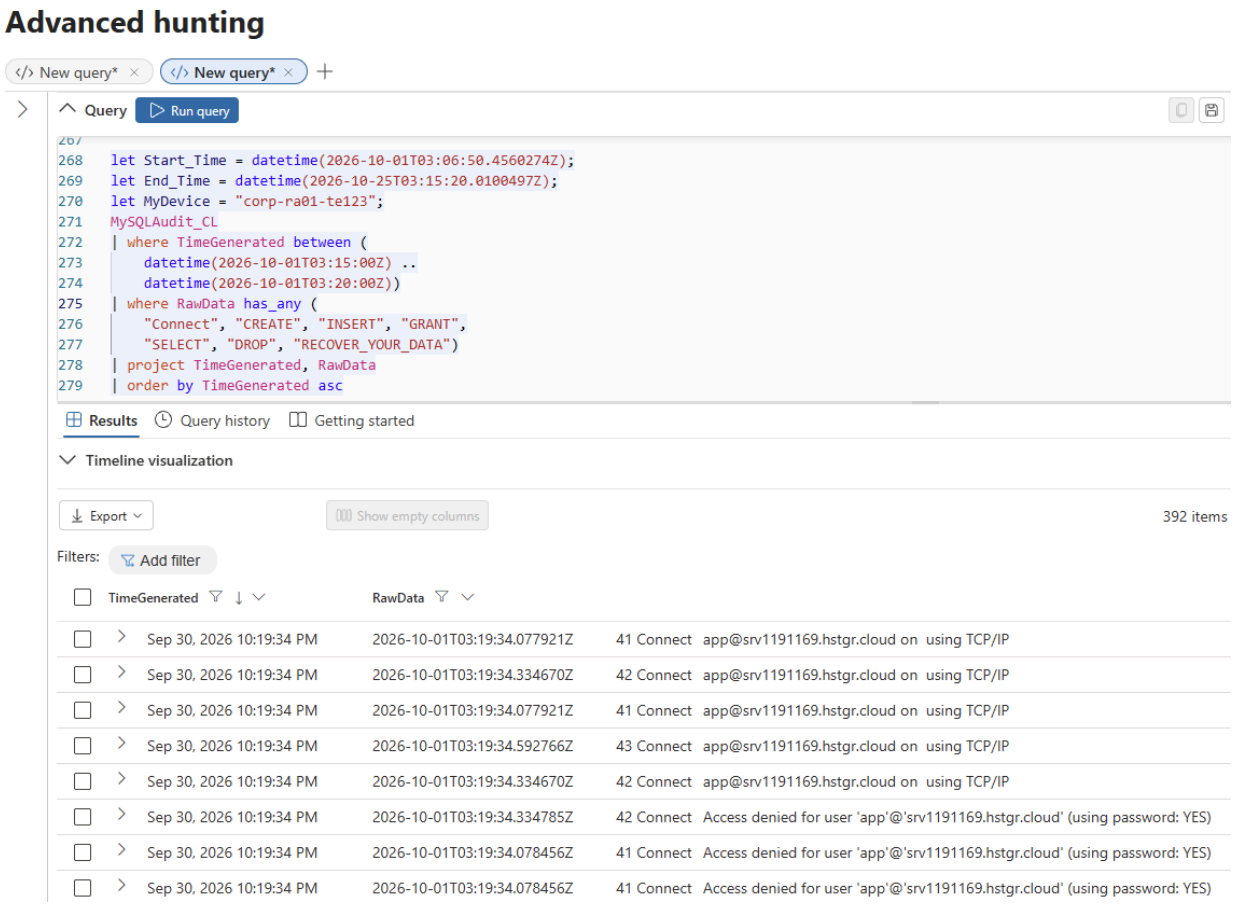
Time	Event	Evidence source

Aug. 30, 2026, 3:12:48 PM (export time)	PowerShell downloads `pwncrypt.ps1` from a GitHub-hosted cyber-range repository.	`Device_Process_Events(2).csv`
Aug. 30, 3:12:52 PM	PowerShell executes `pwncrypt.ps1` using `-ExecutionPolicy Bypass`.	`Device_Process_Events(2).csv`
Sep. 30, 11:12:54 PM	`pwncrypt.ps1` created in `C:\ProgramData`.	`NTANetAnalytics(1).csv`, File
Sep. 30, 11:12:57 PM	PowerShell creates/renames `_pwncrypt.csv` files referencing financial, project, and employee records.	`NTANetAnalytics(1).csv`, File
Oct. 1, 03:16:55 UTC	Ransom database/table creation statements recorded.	`MySQLAudit-CL-Queries_01(2).csv`
Oct. 1, 03:16:56 UTC	Ransom instructions inserted into `RECOVER_YOUR_DATA`.	MySQL query audit
Oct. 1, 03:16:57 UTC	`GRANT CREATE, DROP ON *.* TO root@%` recorded.	MySQL query audit
Oct. 1, 03:16:58–03:17:05 UTC	Sensitive-table queries and database enumeration.	MySQL query audit

Oct. 1, 03:17:33–03:17:34 UTC	Four `DROP DATABASE` statements recorded.	MySQL query audit
Oct. 1, 03:17:35–03:17:37 UTC	Additional ransom-note and privilege-modification statements.	MySQL query audit
Oct. 1, 7:02:50 AM (export time)	MySQL inbound connection accepted from `77.90.185.30`.	`NTANetAnalytics(1).csv`, Network
Unknown	Initial intrusion, SOC alert and formal detection.	Not determined from available logs

KQL hunt — reconstruct MySQL attack sequence

```
MySQLAudit_CL
| where TimeGenerated between (
    datetime(2026-10-01T03:15:00Z) ..
    datetime(2026-10-01T03:20:00Z))
| where RawData contains "DROP DATABASE"
    or RawData contains "RECOVER_YOUR_DATA"
    or RawData contains "GRANT "
    or RawData contains "REVOKE "
| project TimeGenerated, RawData
| order by TimeGenerated asc
```



6. Root Cause / Attack Vector

Confirmed: Privileged MySQL commands were issued, ransom-note content was inserted, and database deletion statements were recorded. Separately, PowerShell execution and `_pwncrypt.csv` file activity occurred on the endpoint.

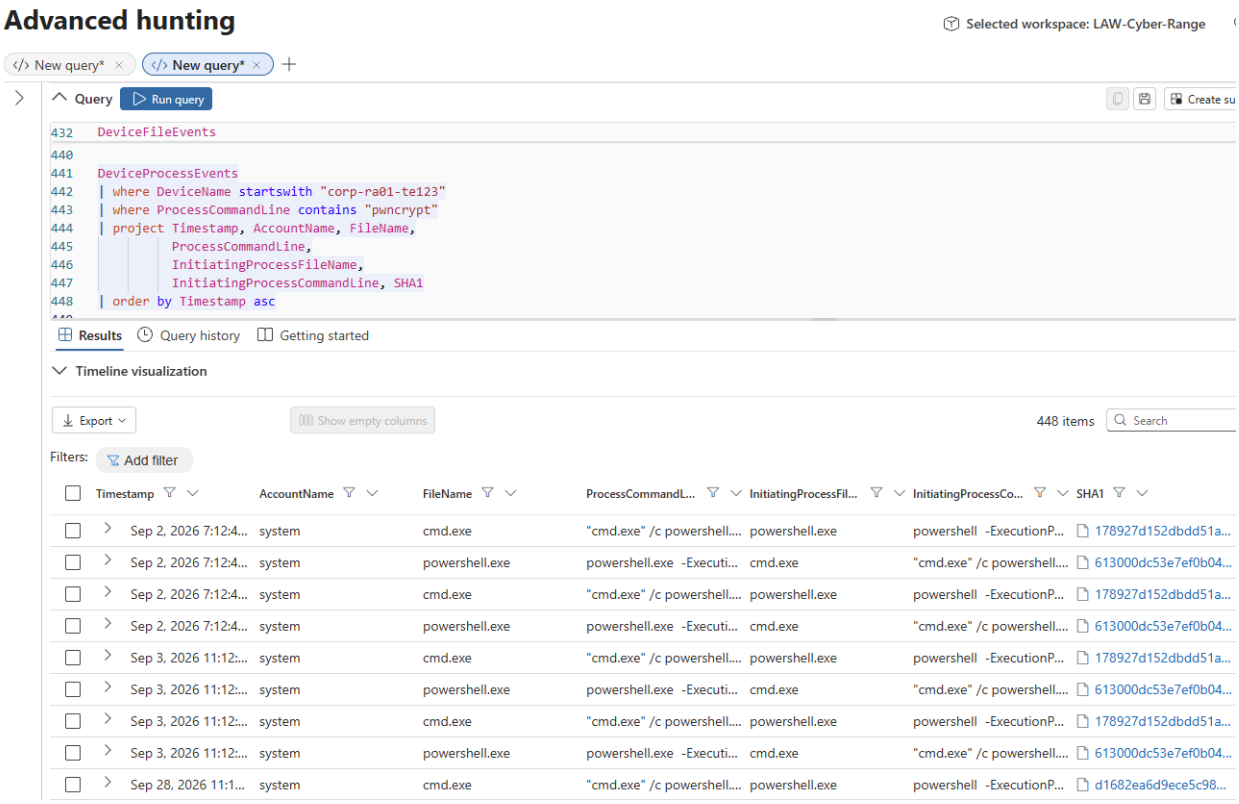
Not determined from available logs: The initial access method, whether credentials were stolen, whether a vulnerability was exploited, and whether the same actor performed the endpoint and database activity.

MITRE ATT&CK technique	ID	Evidence
PowerShell	T1059.001	PowerShell script execution

Ingress Tool Transfer	T1105	Script download
Data from Information Repositories	T1213	Sensitive database SELECT statements
Data Destruction	T1485	DROP DATABASE commands
Valid Accounts	T1078	Possible privileged-account misuse; not confirmed
Data Encrypted for Impact	T1486	Hypothesis based on `_pwncrypt` file operations; encryption not verified

KQL hunt — identify PowerShell command chains

```
DeviceProcessEvents
| where DeviceName startswith "corp-ra01-te123"
| where ProcessCommandLine contains "pwncrypt"
| project Timestamp, AccountName, FileName,
        ProcessCommandLine,
        InitiatingProcessFileName,
        InitiatingProcessCommandLine, SHA1
| order by Timestamp asc
```



7. Response Actions

No completed containment, eradication, or recovery actions are documented in the supplied exports.

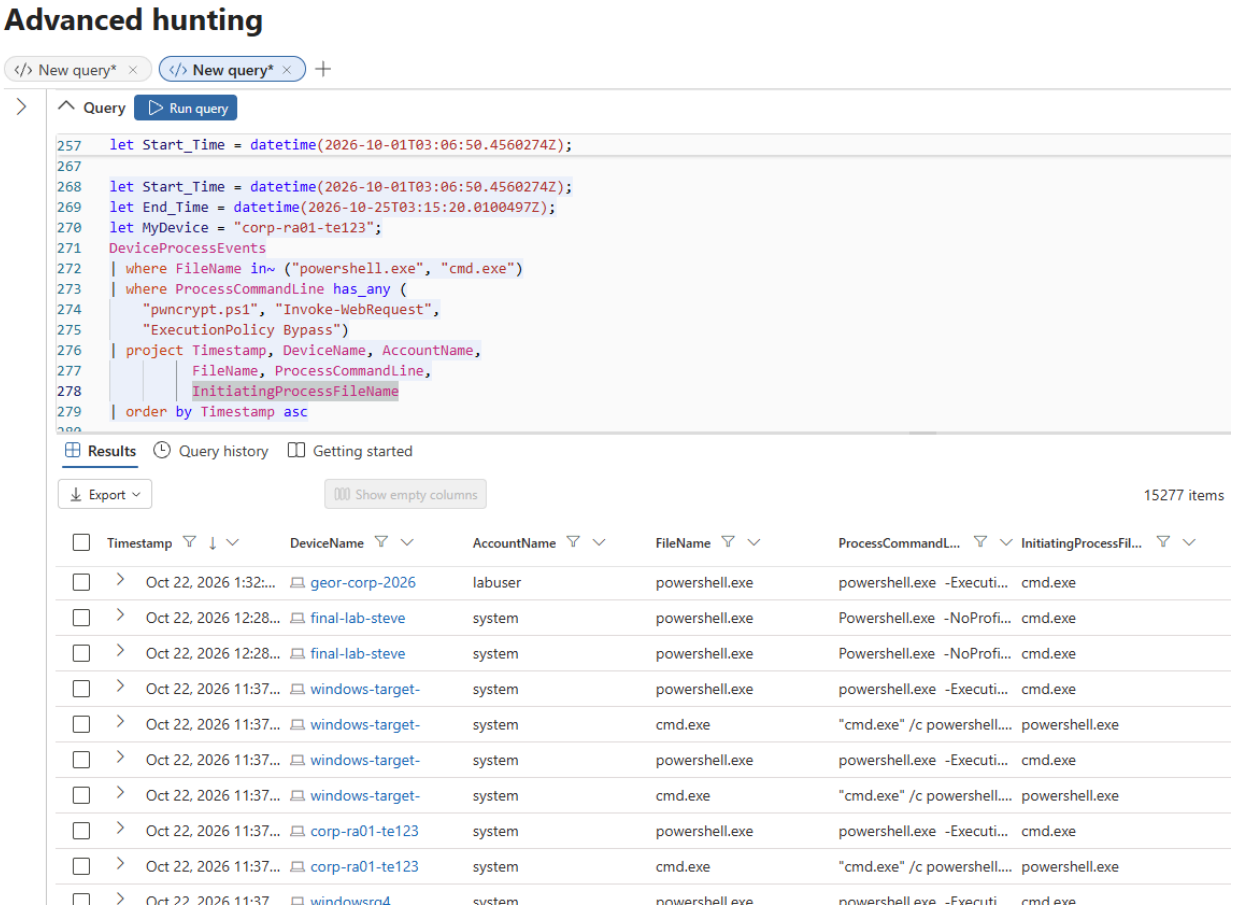
Phase	Recommended response
Identification	Correlate SQL connection IDs, source IPs, PowerShell process trees, and affected files.
Containment	Restrict external MySQL access; isolate affected endpoint if malicious activity remains active.
Eradication	Reset privileged credentials, review MySQL grants, investigate suspicious scripts, scheduled tasks, and registry persistence.

Recovery	Restore affected databases and files from verified clean backups; validate integrity and functionality.
Monitoring	Alert on repeated database authentication failures, destructive SQL, suspicious PowerShell, and unusual outbound traffic.

KQL hunt — identify possible persistence

```
DeviceRegistryEvents
| where DeviceName startswith "corp-ra01-te123"
| where RegistryKey contains @"\CurrentVersion\Run"
   or RegistryKey contains @"\CurrentVersion\RunOnce"
| project Timestamp, ActionType, RegistryKey,
        RegistryValueName, RegistryValueData,
        InitiatingProcessFileName
| order by Timestamp asc
```

Evidence screenshot 3 — PowerShell and persistence investigation



8. Evidence and Forensic Artifacts

The seven uploaded CSVs were reviewed. The two SQL query exports contain the same 837 raw audit records in different schemas and should be treated as one evidence set, not two independent incidents.

Evidence file	Records	Key purpose
`Device_Process_Events(2).csv`	1,144	PowerShell execution and process lineage

`Device_Logon_Events(5).csv`	72	Windows account logons
`MySQLAudit-Auth_01(2).csv`	477	MySQL authentication sources
`MY SQL Audit_01(4).csv`	472	Additional MySQL connection and authentication evidence
`MySQLAudit-CL-Queries_01(2).csv`	837	SQL queries, ransom note and destructive commands
`MySQLAudit-Queries_01(4).csv`	837	Overlapping SQL audit export
`NTANetAnalytics(1).csv`	100,000	Network, file and registry activity

Additional evidence: attempts to disrupt recovery

The MySQL query logs also record the following commands on October 1, 2026:

- `RESET MASTER` — 03:17:36 UTC
- `PURGE BINARY LOGS TO 'josh-mde-lab-bin.000001'` — 03:17:37 UTC
- `REVOKE ALL PRIVILEGES, GRANT OPTION FROM root@%` — 03:17:37 UTC
- `GRANT SHUTDOWN ON *.* TO root@%` — 03:17:37 UTC
- `SHUTDOWN` — 03:17:38 UTC

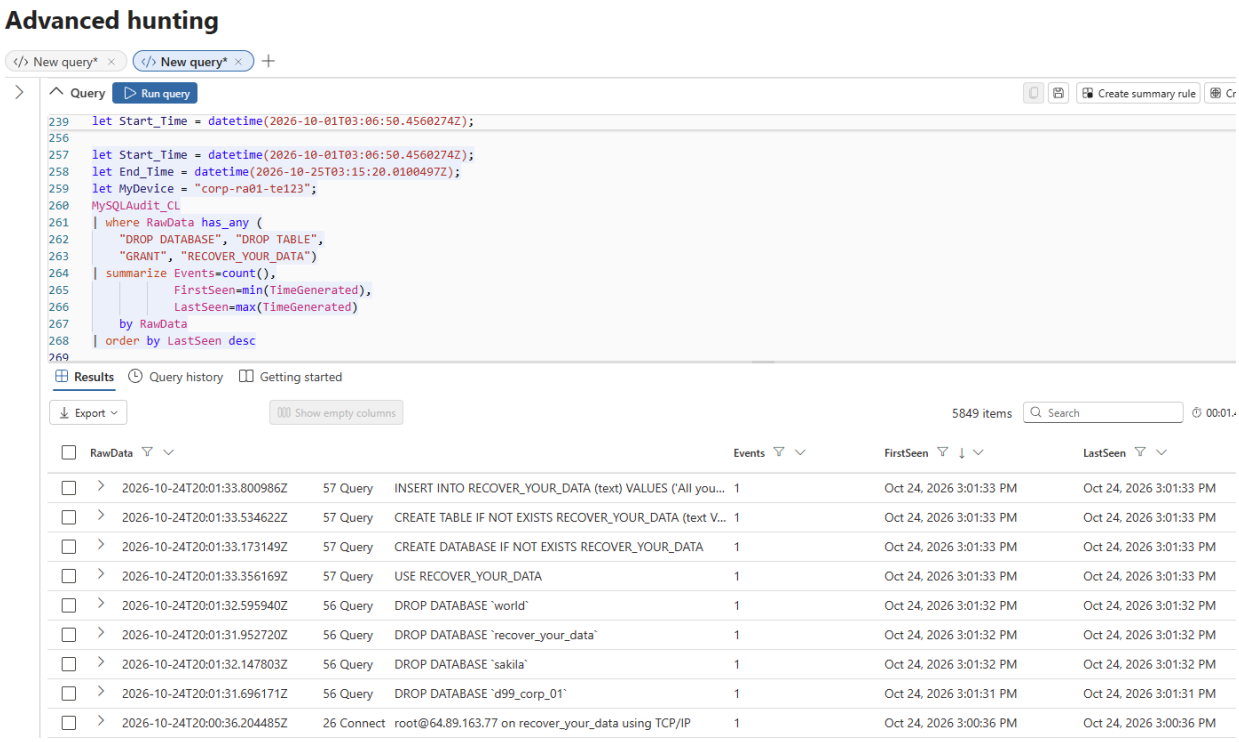
These commands are consistent with attempts to interfere with database operation, logging, or recovery. Their successful execution is not established by the query audit alone.

KQL hunt — database log tampering and shutdown

MySQLAudit_CL

```
| where RawData contains "RESET MASTER"
   or RawData contains "PURGE BINARY LOGS"
   or RawData contains "SHUTDOWN"
   or RawData contains "REVOKE ALL"
| project TimeGenerated, RawData
| order by TimeGenerated asc
```

Evidence Screenshot 4 — Network, File and Registry



Your MDE screenshot shows 771 matching network events, including inbound connections associated with `mysqld.exe` and local port 3306. This supports the presence of MySQL network activity, but does not by itself prove unauthorized database authentication.

For a single combined native MDE hunt, use:

```
let Start_Time = datetime(2026-10-01T03:06:50.4560274Z);
let End_Time = datetime(2026-10-25T03:15:20.0100497Z);
let MyDevice = "corp-ra01-te123";
union
(
   DeviceFileEvents
   | where Timestamp between (Start_Time .. End_Time)
   | where DeviceName startswith MyDevice
   | where FileName contains "pwncrypt"
```

```
    or FolderPath contains "pwncrypt"
  | project Timestamp, DeviceName,
    EvidenceType="File", ActionType,
    Indicator=FolderPath,
    Process=InitiatingProcessFileName
),
(
  DeviceNetworkEvents
  | where Timestamp between (Start_Time .. End_Time)
  | where DeviceName startswith MyDevice
  | where RemoteIP in (
    "77.90.185.30",
    "77.90.185.21",
    "213.209.159.115")
    or LocalPort == 3306
  | project Timestamp, DeviceName,
    EvidenceType="Network", ActionType,
    Indicator=strcat(RemoteIP, ":", tostring(RemotePort)),
    Process=InitiatingProcessFileName
),
(
  DeviceRegistryEvents
  | where Timestamp between (Start_Time .. End_Time)
  | where DeviceName startswith MyDevice
  | where RegistryKey contains @"\CurrentVersion\Run"
    or RegistryValueData contains "pwncrypt"
  | project Timestamp, DeviceName,
    EvidenceType="Registry", ActionType,
    Indicator=RegistryKey,
    Process=InitiatingProcessFileName
)
| order by Timestamp asc
```

Evidence screenshot 4 — Combined MDE hunting results

Paste file, network and registry hunting results here

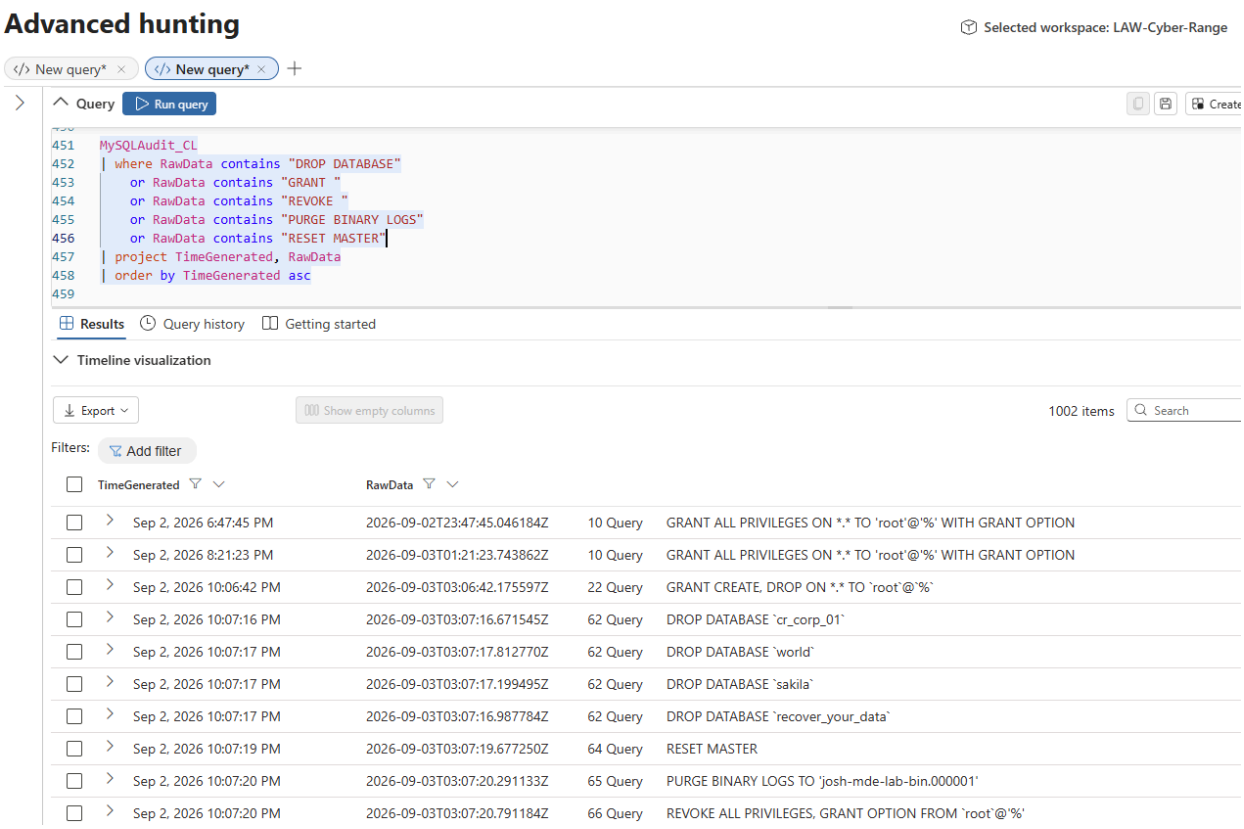
9. Lessons Learned and Recommendations

Priority	Recommendation

P1	Restrict inbound MySQL port 3306 to authorized systems and review internet exposure explicitly.
P1	Rotate privileged MySQL credentials and eliminate unnecessary remote `root` access.
P1	Verify clean, immutable backups, and test the restoration of the four affected databases and the modified endpoint files.
P2	Alert on `DROP DATABASE`, `RESET MASTER`, `PURGE BINARY LOGS`, and unauthorized privilege changes.
P2	Investigate `pwncrypt.ps1` execution, related file operations, and any persistence mechanisms.
P2	Correlate authentication, SQL connection IDs, MDE processes, and network activity using normalized timestamps.

KQL hunt — privileged SQL and destructive operations

```
MySQLAudit_CL
| where RawData contains "DROP DATABASE"
   or RawData contains "GRANT "
   or RawData contains "REVOKE "
   or RawData contains "PURGE BINARY LOGS"
   or RawData contains "RESET MASTER"
| project TimeGenerated, RawData
| order by TimeGenerated asc
```



Final Assessment

Incident type: MySQL destructive database extortion, with suspicious PowerShell-driven endpoint file activity.

Confidence: High that destructive SQL commands and ransom-note operations were recorded. Moderate confidence that endpoint activity represents a related destructive attack, pending validation of file contents and actor correlation.

Not determined from available logs: Initial access, confirmed exfiltration, whether all destructive commands succeeded, total business impact, completed containment, and recovery status.

The most valuable next step is to correlate the SQL session IDs responsible for the ransom and deletion commands with successful MySQL authentication events and source IP addresses. This would strengthen attribution of the destructive database activity to a specific connection.

DFIR Comparative Investigation Report

Evidence: Two Microsoft Defender for Endpoint investigation archives | Host-level forensic comparison

1. Baseline determination

Package	Collection date	Role
`Pre-Breach-MDE_Investigation_Package(1).zip`	October 1, 2026, approximately 01:29–02:17	Baseline
`MDE_Investigation_Package_01(1).zip`	October 25, 2026, approximately 01:10–01:50	Comparison

Collection order is supported by internal ZIP entry timestamps, not simply by the filenames. The baseline contains 349 archive entries; the later package contains 297.

2. Summary verdict

Inconclusive — Further investigation required

Confidence: Moderate. The comparison identifies a security-relevant change to local administrator membership and substantial changes in Prefetch artifacts. These warrant investigation, but the observed differences alone do not prove malicious activity or establish a particular attack type.

The most important finding is that the account `josh` appears in the later local group membership listing but not the baseline. Whether this was an authorized administrative change is not established.

3. Notable artifact deltas

Category	Item	Change	Assessment	Evidence
Local users/groups	`josh` in Administrators	ADDED	Suspicious until authorized	`Users and Groups/LocalGroups.txt`, Administrators section
User profile	`josh` SID ending `-1001`, user Temp directory	ADDED	Requires validation	User-specific uninstall registry export; `Temp Directories/josh_TempDirFiles.txt`
Network services	MySQL TCP 3306 listening on all IPv4 interfaces	UNCHANGED	Exposure risk; not proof of intrusion	`Network Connections/ActiveNetConnections.txt`
Remote access	RDP TCP 3389 listening	UNCHANGED	Requires access-control review	Same network artifact
DNS cache	`scan-81-0.shadowserver.io`	ADDED to snapshot	Ambiguous; not inherently malicious	`Network Connections/DnsCache.txt`
Prefetch	Microsoft Defender and Edge executable entries	ADDED/REMOVED	Likely routine software updates and	`Prefetch Files/Prefetch/`

			Prefetch turnover	
Prefetch	`MYSQLD.EXE` Prefetch no longer present	REMOVED	Ambiguous; MySQL still listening	Prefetch directory and network connections
Installed programs	Machine-wide installed-program export	UNCHANGED	No material installation difference established	`Installed Programs/InstalledPrograms.csv`
OS configuration	Installed KB identifiers changed	CHANGED	Likely normal patching	`System Information/SystemInformation.txt`
Network configuration	Adapter connection name changed	CHANGED	Likely benign configuration change	Same system information artifact

The `josh` finding is a real privilege-membership delta. However, the snapshot does not show who added the account, when membership changed, or whether it was authorized.

4. Strongest indicators and ATT&CK assessment

1. New privileged account membership — highest priority

The later package includes `josh` in the local Administrators group, absent in the baseline. This is consistent with a possible privilege-escalation or persistence mechanism, but also with legitimate account provisioning.

ATT&CK: T1098 — Account Manipulation (potential); T1136.001 — Create Account: Local Account (requires confirmation of account creation).

2. Exposed-service configuration — investigation lead

MySQL port 3306 and RDP port 3389 listen in both snapshots. Listening on `0.0.0.0` does not itself establish internet accessibility or exploitation.

ATT&CK: No technique assigned without evidence of malicious access.

3. Changed executable execution history — low-confidence lead

Prefetch entries differ across the captures, but normal Windows maintenance and Prefetch retention explain many changes. The disappearance of a Prefetch entry does not establish file deletion, anti-forensics, or malicious execution.

ATT&CK: Not assigned.

5. Derived IOC hunt

The following are investigation indicators, not confirmed malicious IOCs. A text search across both packages found no occurrences of `pwncrypt`, `onionmail`, `2mysql`, or `recover_your_data` in the inspected text and CSV artifacts.

Indicator	Baseline	Later	Classification
Account `josh` in Administrators	Absent	Present	High-priority investigation
SID ending `-1001` user registry export	Absent	Present	Account/profile change
`scan-81-0.shadowserver.io` DNS cache entry	Absent	Present	Contextual network indicator

TCP 3306 MySQL listener	Present	Present	Existing exposure
TCP 3389 RDP listener	Present	Present	Existing exposure

No malicious file hash or confirmed command-and-control IP has been established from these differences.

Recommended MDE hunts

Check account creation and administrator membership changes:

```
DeviceEvents
| where DeviceName startswith "corp-ra01-te123"
| where ActionType has_any ("UserAccount", "LocalGroup")
| project Timestamp, DeviceName, ActionType,
        AccountName, InitiatingProcessAccountName,
        AdditionalFields
| order by Timestamp asc
```

Check logons associated with the new account:

```
DeviceLogonEvents
| where AccountName =~ "josh"
| project Timestamp, DeviceName, ActionType,
        LogonType, RemoteIP, AccountName
| order by Timestamp asc
```

For a complete IOC hunt, examine Event IDs 4720, 4732, 4728, 1102, and 104 where the relevant Windows logs exist, and correlate account changes with process and network telemetry.

6. Probable incident type

Undetermined.

The strongest differential finding is a new privileged account membership. This could represent legitimate administration, unauthorized privilege escalation, or persistence. The two packages do not independently establish ransomware, credential theft, lateral movement, or data exfiltration.

7. Evidence gaps

Artifact category	Status
Autoruns and registry persistence	Files present; entry-level differences not fully validated
Running processes and modules	Process exports present; complete semantic comparison outstanding
Services and drivers	Service exports present; detailed differences not fully validated
Scheduled tasks	Exports present; task-level changes not fully validated
Security event logs	Binary EVTX present; event-level comparison and log-clearing assessment outstanding
System/Application event logs	Complete comparable logs not established
File system / Prefetch	Filename comparison completed; binary Prefetch execution metadata not decoded
Hosts file / firewall rules	Complete rule-level and hosts-file comparison not established
Browser / credentials	Sufficient comparable evidence not established

File hashes	No verified suspicious executable hashes extracted

These gaps limit the confidence of the verdict; absence from a snapshot or text search is not proof that activity never occurred.

Final DFIR Assessment

The October 25 capture shows a meaningful change to local administrator membership compared with October 1. That is the highest-priority finding, but compromise remains inconclusive.

The immediate next step is to determine when `josh` was created or added to Administrators, which account performed the change, and whether an approved administrative action explains it. A complete EVTX, Autoruns, scheduled-task, and service comparison is also required before closing the investigation.